

CSE 310 Online (A2): Yet Another Tag Matching Lex Problem

Time: 40 minutes

1 Objective

In this task, you will create a new .1 file, which must not be reused from your offline laboratory assignments, to validate the proper nesting of tags in a simplified XML-like document.

Each opening tag must have a corresponding closing tag, and the tags must be closed in the correct order.

2 Supported Tags

Only the following tags are supported:

- `<book>` and `</book>`
- `<title>` and `</title>`
- `<author>` and `</author>`

3 Requirements

- Detect and validate the supported opening and closing tags.
- Each closing tag must match the most recently encountered unmatched opening tag.
- One nice way is to maintain a stack to manage the nesting of tags. [stack demo code](#)
- Ignore all text content between tags. For example, for the following input, the text `Compiler Design` should be ignored:

`<title>Compiler Design</title>`

- Any tag other than `book`, `title`, and `author` must be reported as unsupported.
- If a closing tag does not match the most recent opening tag, report a tag mismatch.
- If a closing tag appears without a corresponding opening tag, report it as an unexpected closing tag.
- If one or more opening tags remain unmatched at the end of the input, report the missing closing tag.
- Stop parsing immediately after detecting the first error.

4 Sample Input and Output

Sample Input 1

```
<book>
<title>Compiler Design</title>
<author>A. Aho</author>
</book>
```

Expected Output

Valid XML structure

Sample Input 2

```
<book>
<title>Compiler Design</author>
</book>
```

Expected Output

Invalid XML: tag mismatch </author>

The parser should stop when </author> is encountered because the most recent unmatched opening tag is <title>.

Sample Input 3

```
<book>
<title>Compiler Design</title>
```

Expected Output

Invalid XML: missing closing tag for <book>

Sample Input 4

```
<book>
<publisher>ABC Press</publisher>
</book>
```

Expected Output

```
Invalid XML: unsupported tag <publisher>
```

Sample Input 5

```
<book>  
</author>  
</book>
```

Expected Output

```
Invalid XML: tag mismatch </author>
```

Sample Input 6

```
<book>  
<title>Compiler Design</title>  
<author>A. Aho</author>
```

Expected Output

```
Invalid XML: missing closing tag for <book>
```

5 Marks Distribution

- Correct stack-based matching and nesting of tags: **5 marks**
- Detecting the listed errors: **5 marks**

6 Submission Instructions

- Submit only your new Lex file.
- Name the file as 22XXXXX.1.